

Chapter 66. The Ethics of Withholding and Withdrawing Mechanical Ventilation

Michael E. Wilson; Elie Azoulay

The Ethics of Withholding and Withdrawing Mechanical Ventilation: Introduction

To save the lives of critically ill patients, intensivists utilize sophisticated technologies to support vital organs until treatments reverse underlying medical conditions. Most patients recover from the acute event, a few die rapidly, and the remainder fail to improve and remain dependent on life-sustaining treatments. In this last group, the chance of recovery changes from one day to the next, and questions often arise about the appropriateness of continuing life support, especially mechanical ventilation.¹

Over the last half century, health care professionals in intensive care units (ICUs) have been forced to make decisions for patients who remain dependent on mechanical ventilation with death in the short term as the only possible outcome.² In these patients, continued treatment in the hope of cure is rarely the best option.³ Mechanical ventilation may be prolonged beyond the point of beneficence, robbing patients of their dignity and families of their right to honest prognostic information and an opportunity to prepare for bereavement. The best option here is a decision to forego life-sustaining treatment.

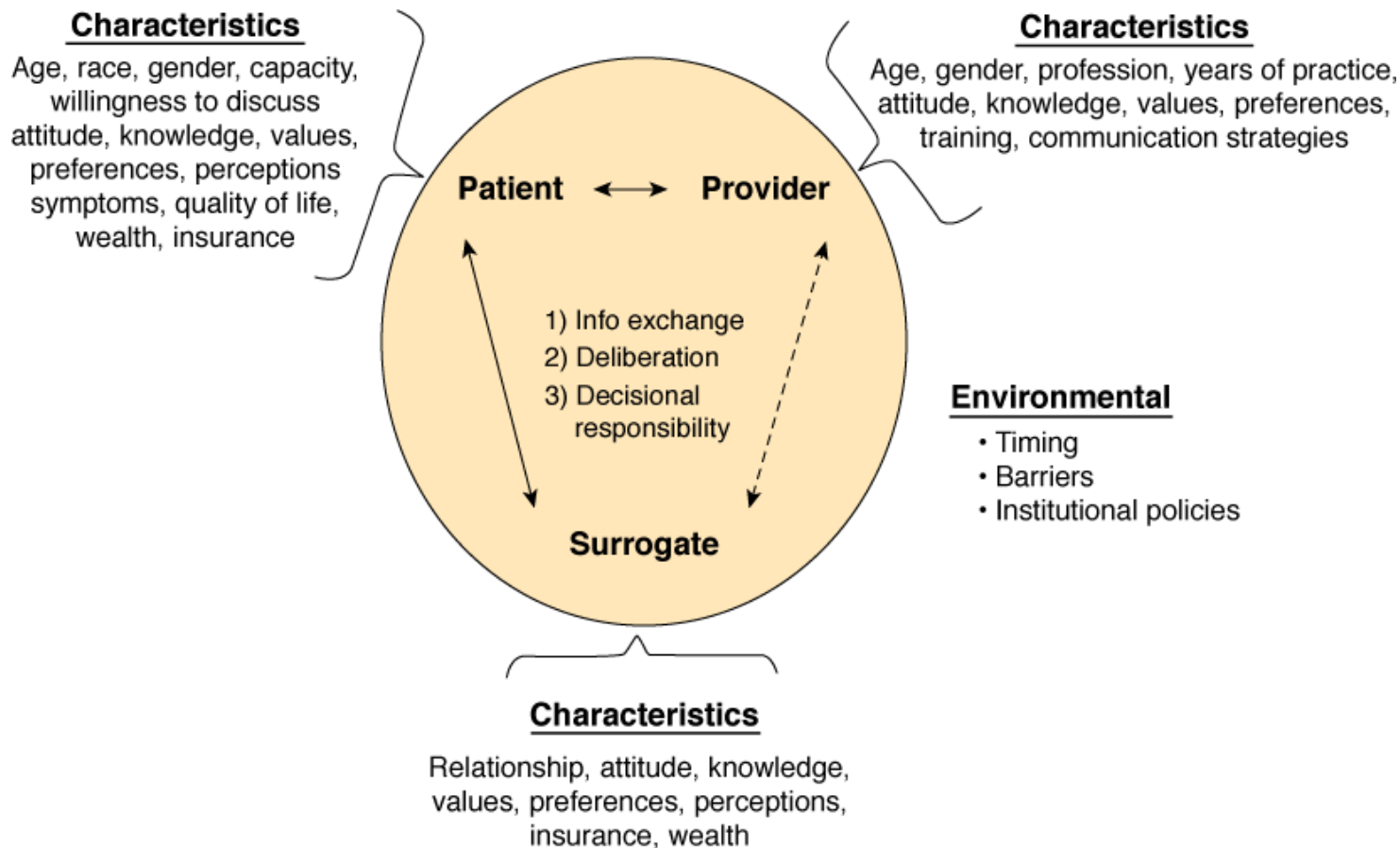
Because respiratory failure, shock, and coma are common reasons for ICU admission, mechanical ventilation is the most widely used life-sustaining treatment in the ICU.⁴ Thus, mechanical ventilation is also the most common target of a decision to forego life-sustaining treatment.⁵⁻¹⁰ Although most patients are successfully weaned off the ventilator, a few die while on the ventilator or immediately after weaning.¹¹ Ideally, a decision to forego life-sustaining treatment, which consists of moving from curative care to comfort care, should be based on the patient's wishes.^{12,13} When the issue of comfort care arises, however, fewer than 5% of patients are able to participate in decisions, and knowledge of their preferences is usually unavailable.^{14,15} Therefore, concern that curative care may be harmful is often voiced first by the ICU team, which then broaches the issue with the family or surrogate decision maker. Thus, barely a few years after the creation of ICUs, intensivists realized that, in addition to fighting death, their duties included the daunting task of accepting and managing death. This task requires (a) identifying situations in which all hope of recovery is lost and life-prolonging treatments become death-prolonging treatments, which should be withdrawn or withheld; (b) promptly initiating a continuous process of family care based on sensitive and straightforward information and communication; and (c) improving the ability to manage death, via epidemiologic studies of practices, interventional studies of end-of-life strategies, and continuing education aimed at honing the information and communication skills of all ICU professionals. Warding off death and restoring self-sufficiency have been the main goals of intensivists for decades; now, ICU professionals are becoming acutely aware that they must develop a professional approach to dying patients, learn what makes a "good death," and provide dying patients and their families with support, reassurance, comfort, dignity, and freedom from guilt.

The literature on end of life in the ICU comprises epidemiologic studies (descriptive, deductive, or quantitative) and qualitative studies of theoretical concepts that allow subtle interpretations of structures, experiences, roles, interactions, and perspectives. This review provides an interpretation of published data on limiting mechanical ventilation and other life-supporting treatments in patients dying in the ICU. Its goal is to help readers understand and organize the decision-making process within an ICU team and to ensure that decisions are implemented so as to give the patient a “good death” and families bereavement support.

Fundamental Issues of Ethics

The field of bioethics, born in the late 1960s, rests on four fundamental ethical principles (beneficence, nonmaleficence, autonomy, and distributive justice) and describes a spectrum of patient–physician relationship styles ranging from paternalism to autonomy (Fig. 66-1).¹⁶ The four fundamental ethical principles guide decisions in the ICU management of dying patients and their families. Regarding beneficence and nonmaleficence, we discuss studies that address the specific needs and expectations of dying patients. We then briefly contrast paternalism and autonomy, and argue that the long-standing controversy opposing these two models should give way to emphasis on the shared decision-making model. We also discuss the double-effect principle accepted by the U.S. Supreme Court, Society of Critical Care Medicine recommendations, and international consensus in the support of the use of sedation and analgesia to relieve symptoms, provided death is not intended, although it may be foreseen.^{17–20} The unresolved debate regarding the relative merits of terminal extubation and terminal weaning for taking dying patients off the ventilator is mentioned. Regarding justice, we do our utmost to convince the reader that cost considerations are irrelevant to a decision to forego life-sustaining treatment.

Figure 66-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

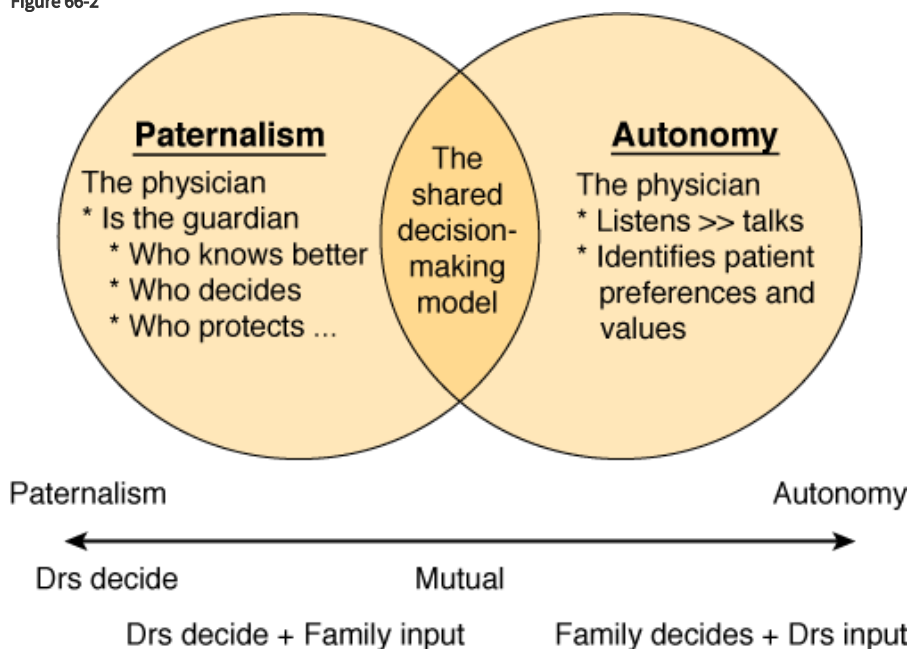
Conceptual framework of a patient-physician interaction. (Adapted, with permission, from Heyland DK, Rocker GM, Dodek PM, et al. Family satisfaction with care in the intensive care unit: results of a multiple center study. *Crit Care Med*. 2002;30:1413-1418.)

In a study by Singer et al,²¹ patients identified five domains of quality end-of-life care: adequate relief from pain and anxiety,²² avoiding inappropriate prolongation of dying; achieving a sense of control; relieving burden; and strengthening relationships with loved ones. When a patient is dying, intensivists must make it clear that they are dedicated to providing optimal care throughout the dying process;²³ to treating the patient with respect and dignity; and to relieving pain caused by physical, emotional, social, and spiritual factors.^{24,25} Patients fear to be abandoned. They should be assured that the doctor is and will remain on their side, stopping useless interventions, and providing treatments that ensure comfort.²⁶ Encouraging family and friends to be present at all times is another component of this effort to ensure beneficence; however, some family members may be unbearably distressed by having to spend long hours with their dying relative. The presence of a chaplain, chosen by the patient and family, and access to religious rites should be encouraged.²⁷

After years of heated debate opposing autonomy and paternalism, a model in which decision making is shared with family members is gaining precedence (Fig. 66-2). This model upholds patient autonomy²⁸ without forcing family members to be involved in decisions they do not want to make²⁹ or are not ready to make.³⁰ The shared decision-making model stands in sharp contrast to paternalism, in which the physician shields the patient, making decisions alone so as to protect the patient and family from the potentially harmful effects of making painful decisions.¹³ Because most ICU patients are unable to make decisions,^{31,32} sharing in a decision to forego life-sustaining treatment shifts to the family members.¹²

Attempting to wake ICU patients so that they can participate in decisions in the name of autonomy clashes violently with the principles of beneficence and nonmaleficence.³³ Beneficence requires that family members be empowered to understand the patient's situation,³⁴ to identify and meet their expectations,^{35,36} and to gain awareness of possible anxiety or depression that might impair their decision-making capabilities.³⁷ Under no circumstance should anxiety or depression in family members be used to justify benevolent paternalism; on the contrary, communication with families must receive close attention as a means of empowering families to share in decisions.¹² Reports from Canada, Sweden, and the United States describe sharing discussions and decisions as rational³⁸ and as crucial to family satisfaction.^{39–42} In addition, studies of family outcomes several weeks or months after the death of a patient in the ICU have highlighted major difficulties and profound inadequacies in information.^{27,43–45} Finally, nonmaleficence in this setting requires intensive communication with families if needed; a multidisciplinary approach can be used, or external ethical advice obtained, with the objective of empowering families to achieve their own goal (whether this is sharing in decisions or leaving decisions to the intensivists) and of convincing families that comfort care is preferable over aggressive interventions.^{46–48} When family members have not received optimal information, involving them in the decision-making process probably carries a risk of subsequent posttraumatic stress and abnormal grief reactions.³⁰

Figure 66-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

The shared decision-making model.

The SUPPORT (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) studies showed a high rate of unacceptable pain in dying patients.^{49,50} When withholding or withdrawing mechanical ventilation is in order, clinicians have the duty to emphasize comfort and to relieve pain and anxiety in their patients. Opioids and anxiolytics remain the reference treatment for these symptoms.^{17–20} Opioids may hasten death by inducing respiratory depression,⁵¹ but this risk is legally and ethically acceptable.^{19,52–54} In contrast, use of high-dose opioids to cause death by a person making the decision alone is not consonant with optimal end-of-life care. This is voluntary euthanasia and is illegal, even in The Netherlands, Belgium, and the states of Oregon and Washington, where patients can request and obtain the assistance of a physician to commit suicide.^{19,52} Because it is the physician's intention that separates opioid use to

relieve pain and anxiety from opioid use to hasten death, the line between the ethical and the unethical, the legal and the illegal, is subjective.^{18,55} Although some physicians may tend to use higher doses to ensure patient comfort, this practice cannot be likened to euthanasia.⁵⁶ Nevertheless, eliminating ambiguity from end-of-life decisions can be extremely difficult.⁵⁷

The optimal method for taking dying patients off the ventilator remains actively debated. The controversy consists of advocates of terminal weaning, in which volumes, respiratory rate, and fractional inspired oxygen concentration (FI_{O_2}) are reduced gradually, versus proponents of terminal extubation with sedation,⁵⁸ which restores the normal appearance of the patient but carries a risk of respiratory secretion accumulation with asphyxia and gasping.^{59,60} Terminal weaning is perceived as less active and therefore less distressing for health care professionals; furthermore, the family members are spared the ordeal of witnessing gasps, which they interpret as suffering.⁶¹ In addition, extubation may wrongly suggest to the family that the patient is better and no longer needs the ventilator.⁶¹ Opponents of extubation for ventilation withdrawal have pointed out that even patients who survive extubation remember the weaning period as a time of stress, discomfort, loss of hope, and extreme fear.⁶²

Cost considerations⁶³ have entered into the treatment-limitation debate on the grounds that using ineffective treatments in one patient may deprive another patient of lifesaving treatment or ICU admission, thereby violating the principle of distributive justice.⁶⁴ Guidelines⁶⁵ suggest that elderly patients and patients with chronic conditions (cancer, hematologic malignancies, or chronic obstructive pulmonary disease) should be denied ICU admission or should receive a decision to forego life-sustaining treatment earlier than other patients. Whether the principle of distributive justice applies to a decision to forego life-sustaining treatment is controversial, and the relevance of the cost/efficacy concept to ICU management has been challenged.⁶⁶ Denying ICU admission to a seriously ill patient for cost-containment reasons makes little sense, because other resources will then be used for the patient elsewhere, sometimes for a longer period and to the detriment of patient comfort.⁶⁷ Furthermore, patient-centered care allows the patient to die with dignity, free of pain and anxiety, while placing the family in optimal conditions for preparing the bereavement process.^{64,68,69} Interventional studies of intensive communication with families of patients dying in the ICU found cost savings related to conflict prevention^{70,71} that diminished the use of ineffective treatments.^{46–48,72}

Terminology

Five mutually exclusive categories of a decision to forego life-sustaining treatment have been defined. The first category is the do-not-resuscitate order for cardiac arrest. The second category comprises a decision to forego life-sustaining treatment in patients who are in a chronic vegetative state. Third is the decision to withhold a potentially beneficial treatment (e.g., mechanical ventilation, catecholamine administration, or dialysis). Fourth is withdrawal of life-sustaining treatment. The fifth category involves active induction of death (administration of a treatment that results in death). Do-not-resuscitate orders are not relevant to the present chapter. We do not discuss patients in a chronic vegetative state, because they raise specific issues whose resolution relies on concepts that do not apply to other ICU patients. Neither do we deal with euthanasia (injection of a lethal substance), which has been used in the past but is no longer an acceptable component of end-of-life care in ICU patients. Euthanasia does not constitute care, virtually never involves a rational collegial decision-making process, leaves anxiety and pain unrelieved, and fails to respect the dignity of the patient and family. We believe that euthanasia should be banned from ICUs and all other places where medical care is delivered.

Epidemiology of the Decision to Forego Life-Sustaining Treatment in the Intensive Care Unit: Geographic, Cultural, and Religious Variations

In most countries, most deaths in the ICU are preceded by a decision to forego life-sustaining treatment.^{6,7,9,40,56,73–81} In addition, the incidence of decisions to forego life-sustaining treatment may be increasing over time.^{82,83} In patients discharged alive from the ICU, a decision to forego life-sustaining treatment taken in the ICU influences hospital survival.⁸⁴ In practice, withholding mechanical ventilation precedes or occurs concomitantly with withdrawal of all other life-sustaining treatments.^{10,73} Withholding and withdrawal occur within 3 days after ICU admission, and the patient usually dies within the next 24 hours.⁸⁵ There is widespread agreement that there is no ethical difference between withholding and withdrawal,⁸⁶ although withdrawal has been described as more difficult for intensivists⁷⁴ and is not used in some countries.⁷³

Wide variation exists regarding the incidence of decisions to forego life-sustaining treatment (Table 66-1).⁸⁷ These variations across countries and cultures have been identified in studies comparing practices,^{88,89} studies of responses to ethical scenarios,^{74,90} and descriptive studies.^{7,9,10,73,75,78,82,91–105} The Ethicus study described the variable practices to forego life-sustaining treatments in thirty-seven ICUs in seventeen European countries (Table 66-2). In southern Europe, the proportion of deaths after unsuccessful resuscitation was greater than in the rest of Europe. In northern Europe, the time from a decision to forego life-sustaining treatment to death was shorter.⁷³ These data confirm the considerable variability in approaches to bioethical issues in Europe.^{88,106}

Table 66-1: Variations Across Countries in the Incidence of Decisions to Forego Life-Sustaining Treatments and in Family Involvement in These Decisions

	References	WH/WD in Dying Patients (%)	Involvement of Families in Discussions and Decisions (%)
United States	6, 11, 75, 78, 82, 102	>90	90
Canada	94	80	90
Europe	73	76	—
Israel	79	91 (no WD)	28
England	102	85	>90
France	7, 9, 30	50	50
Hong Kong	126	23 to 61	95
Spain	76	34	41
Italy	98	8	58
Australia	101, 105	70 to 81	—
India	103	49	100

Abbreviations: WD, withdrawal; WH, withholding.

Table 66-2: Distribution of Categories of Dying Patients in Europe

	Unsuccessful CPR	Brain Death	WH	WD	Active Shortening of the Dying Process
Northern Europe	154 (10.2%)	48 (3.2%)	575 (38.2%)	714 (47.4)	14 (0.9%)
Central Europe	217 (17.9%)	92 (7.6%)	412 (34.1%)	409 (33.8%)	79 (6.5%)
Southern Europe	461 (30.1%)	190 (12.4%)	607 (39.6%)	275 (17.9%)	1 (0.1%)
Whole of Europe	832 (19.6%)	330 (7.8%)	1594 (37.5%) 1	1398 (32.9%)	94 (2.2%)
Hospital mortality	100%	100%	89%	89%	100%
Total	832 (19.6%)	330 (7.8%)	1594 (37.5%)	1398 (32.9%)	94 (2.2%)

Abbreviations: CPR, cardiopulmonary resuscitation; WD, withdrawal; WH, withholding.

Source: Adapted, with permission, from Sprung et al.⁷³

Additionally, the role given to the patient's and family's opinion in the decision to forego life-sustaining treatment also varies widely, although compassion and respect for the patient are universally recognized as crucial.^{12,13} Some differences may be explained by the fact that in the late 1980s the United States adopted a model based on patient autonomy and self-determination,^{96,107} whereas other countries kept a paternalistic model in which the physician alone determines the appropriate level of treatment intensity.^{7,9,74,90,91,108,109} Evidence, however, suggests that variability exists within a given country, with some physicians in traditionally paternalistic countries involving patients and families in the life support decision making process, and vice versa.^{75,78,82,94}

Religious beliefs of both patients and providers are reported to influence the attitudes of intensivists towards life support decisions.^{88,110,111} The Ethicus study identified variations across religions with Jewish, Greek Orthodox, and Moslem physicians more likely to withhold therapy, and Catholic and Protestant physicians more likely to withdraw therapy. Time to first therapy limitation also varied from 1.6 days for Protestant physicians to 7.6 days for Greek Orthodox physicians.^{73,112}

Variable ICU admission rates and policies may influence the variable incidence of decisions to forego life-sustaining treatment. The SUPPORT study group found that half the patients with chronic diseases were in the ICU within the 3 days preceding death and that one-third of patients spent at least 10 days in the ICU during the hospitalization that preceded death.⁵⁰ In 1995, 20% of deaths in the United States or Canada occurred in the ICU.^{113,114} In other countries, ICU admission policies are more restrictive,^{102,115–117} so that comparisons are inherently biased. Predictors of increased

end-of-life ICU utilization include age, number of comorbidities, hospital type, number of available hospital beds, and lack of outpatient to inpatient continuity of care.^{118–120} The high incidence of in-hospital and in-ICU deaths, which has emerged over recent decades in industrialized countries, must be considered under the harsh light of data showing poorer quality end-of-life care in hospitals and ICUs as compared to home hospice care,^{121,122} in addition to minimal gains in survival.¹²³ Important differences also exist among race. In the United States, black and Hispanic patients are more likely to die in an ICU¹²⁴ and incur 30% to 50% more health care expenditures in the 6 months preceding death than white patients,¹²⁵ even after control for variables such as geography and socioeconomic status. Additionally, there is limited evidence that white patients who prefer intensive end-of-life care are more likely to receive it than are black patients with the same preference.¹¹¹

Legal Decisions to Date

The courts recognize that withholding or withdrawing life-sustaining treatment and giving palliative care are legal. In the United States, the ethical principle informing laws that allow decisions to forego life-sustaining treatment is autonomy, manifesting as informed consent from the patient or family, who can also refuse treatment withholding or withdrawal.^{14,19,52,126} Among European countries, The Netherlands was the first to allow euthanasia and physician-assisted suicide.¹⁰⁶ Belgium followed suit in 2002.^{127,128} Nevertheless, in neither country does the law deal specifically with patients receiving life support in the ICU.¹²⁹ In France, the Senate has passed a law authorizing physicians to let patients die if they are kept alive only through artificial means, treatment is futile, and death is imminent.¹³⁰

After the advances in mechanical ventilation achieved in the 1970s and 1980s, the notion that mechanical ventilation should at times be withheld or withdrawn was deeply disturbing to many intensivists. By the 1980s, however, intensivists could no longer deny that life-sustaining treatment merely prolongs the dying process in some patients.^{22,86,107}

The confusion that exists between ethical concepts and legal concepts is frequently disconcerting to intensivists. From both the ethical and legal points of view, a decision to forego life-sustaining treatment is acceptable only when it constitutes an expression of the patient's personal autonomy, that is, when it is made with the informed consent of a competent patient or, when the patient is incompetent, based on knowledge of the patient's wishes. All adults have the right to accept or refuse treatment and to define their preferences and values.^{28,33,131,132} This right does not end when an ICU patient becomes mentally incompetent, and it can be exercised by the patient's surrogate decision maker.^{14,52} In 1990, the U.S. Supreme Court upheld the withdrawal of life-sustaining treatment at a patient's request,¹³¹ more recently, it stated that decisions to forego life-sustaining treatment in ICUs did not constitute physician-assisted suicide or euthanasia.¹³³ These rulings apply to all fifty states, in theory at least. Furthermore, the Supreme Court has ruled that physicians cannot use futility as a basis for making a decision to forego life-sustaining treatment of their own accord. Nevertheless, the Supreme Court issued detailed recommendations on palliative care, acknowledging that sedatives and analgesics may be given, when needed, to alleviate the symptoms of a dying patient, even when this is expected—but not intended—to hasten death (the doctrine of double effect).¹⁹

The information above relates to the law. Physicians must comply with the laws of their country. A medical decision, however, may be legal yet unethical. We believe that the confusion between ethical principles and legal obligations that exists in the minds of many physicians can distort the decision-making process. When decisions to forego life-sustaining treatment are entirely based on standardized criteria, which comply with the law but ignore the specific factors

characterizing each individual patient,¹³⁴ they may produce deleterious effects, ranging from loss of opportunity to treat to administration of useless treatments. Furthermore, the confusion between ethical and legal obligations may hinder openness in communicating decisions; for instance, physicians may be reluctant to record the nature and implementation of a decision to forego life-sustaining treatment in the patient's medical record.¹³⁵

When decisions to forego life-sustaining treatment are made, fewer than 5% of patients are able to participate in the discussions or decisions.¹³⁶ In addition, patient preferences are usually not known.²⁰ Consequently, intensivists have turned to families as the primary partners for initiating a decision to forego life-sustaining treatment. The patient may be represented by a surrogate decision maker holding a durable power of attorney for health care; if the patient has not designated a surrogate, the intensivists discuss decisions with family members. The person representing the patient is asked for advice in some countries^{130,137} and for decision-making input in others.^{19,138} Whether the information on patient wishes used to make a decision to forego life-sustaining treatment in patients receiving mechanical ventilation should be obtained from family members has been challenged. Many families do not understand what is at stake;¹³⁹ have no knowledge of the patient's wishes;¹⁴⁰ want, but do not have, written instructions from the patient;¹⁴¹ have opinions that disagree with patient wishes;¹⁴² and, most importantly, suffer a burden of stress and anxiety that may impair their decision-making capacities.³⁷

Advance directives have been suggested as a means of allowing patients to remain in control of their care, even when they have lost their decision-making capacity, as is the case for intubated patients in the ICU.¹⁴³ Unfortunately, a complex set of reasons¹⁴⁴ impairs the effectiveness of advance directives in ensuring that patient wishes are honored¹⁴⁵ and that treatments given in the ICU or wards are changed in accordance with patient instructions.^{68,146} In addition, patients may change their minds over time, so that advance directives may no longer reflect their wishes at the time decisions to forego life-sustaining treatment are made.^{68,144–147} Interestingly, a 1990 survey among members of the American Thoracic Society found that one-third of respondents provided care that contradicted the wishes of patients and surrogates, and that more than 80% unilaterally made and implemented a decision to forego life-sustaining treatment, at times over the objections of patients and surrogates.⁹² Finally, the SUPPORT investigators found that 10% of older inpatients with serious illnesses received care they did not want⁴⁹ and that advance directives or families' wishes were often disregarded.⁵⁰ Having a specifically trained nurse talk to the patients, families, physicians, and other hospital staff members failed to improve compliance with patients' wishes.⁵⁰

The Decision-Making Procedure: High Complexity, High Stakes

Evaluating the prognosis is the key issue: When continuing mechanical ventilation and other aggressive interventions that may ensure recovery from the acute life-threatening event that prompted ICU admission, full care must be provided, without which the patient may suffer a loss of chance. Therefore, intensivists must be able to identify those patients whose chances of recovery are virtually nonexistent. For these patients, aggressive care, far from inducing benefits, prolongs the dying process, puts the patient's dignity in jeopardy, and hinders the bereavement process for family members. Clearly, the stakes are extraordinarily high for all concerned with the decision, who must achieve a consensus about what to do and how to do it.

Because decisions to forego life-sustaining treatment force patients, families, and health care professionals to stand very close to the line that separates killing from allowing patients to die, they carry a destructive power that must be acknowledged and kept under control. To this end, intensivists have worked on developing decision-making procedures

that maximize objectivity, legitimacy, serenity, and agreement among all those involved. Two main factors govern a decision to forego life-sustaining treatment: the imminence of death and the patient's wishes regarding life support. The first factor was investigated by pursuing two avenues concomitantly: one consisted in defining futility, a concept based on clinical experience, and the other in developing mortality-prediction tools based on physiologic disturbances or organ failures. These efforts were intended to assist in objectively identifying situations that warrant a decision to forego life-sustaining treatment; however, they failed to reduce the complexity of the decision-making process. The second factor, patient wishes, faced a major obstacle: The inability of most patients to express their wishes at the time a decision to forego life-sustaining treatment is considered.^{10,20,73} To ensure that patient wishes would nevertheless be honored, three approaches were suggested: advance directives, formal designation of a surrogate decision maker holding a durable power of attorney, and family participation in decisions.

Predicting mortality has relied chiefly on severity scores developed in the United States and Europe^{148–151} to characterize patients (based on age, chronic morbidities, recruitment type, and severity of each organ involvement) and to constitute homogeneous patient groups (i.e., groups with a similar risk of hospital death). These scores have proved useful for measuring the performance of ICUs (via determination of the standardized mortality ratio [SMR]) and for establishing homogeneous patient groups for inclusion in therapeutic trials. Their poor calibration and discrimination, however, make them unhelpful for predicting the risk of death in the individual patient. Furthermore, because these scores are determined at a single point in time (ICU admission) and developed in a given population, they fail to reflect changes in organ failures over time^{152,153} or advances made in the management of a specific condition.¹⁵⁴ Therefore, mortality-prediction scores are of no assistance in deciding when treatment withholding or withdrawal is appropriate.

Futility is a concept that was widely used to determine that a decision to forego life-sustaining treatment was appropriate. The patient-benefit-centered view defines futility as use of interventions that are unlikely to benefit the patient.¹⁵⁵ We prefer the definition suggested by Schneiderman et al in 1990: “When physicians conclude (either through personal experience, experiences shared with colleagues, or consideration of published empiric data) that in the last 100 cases a medical treatment has been useless, they should regard that treatment as futile,” and “if the likelihood of functional recovery after a proposed course of therapy is less than 1%, then physicians may assert the prerogative to withdraw therapy without the consent of the patient or surrogate decision maker.”¹⁵⁶

The concept of futility has been fiercely criticized. Its opponents argue that the definition of futility is neither clear nor reproducible, the concept is clearly intended to increase the power of the physician while undermining patient autonomy, treatment data do not necessarily apply to an individual patient, the 1% cutoff is not supported by scientific evidence, and the use of futility allows physicians to make decisions unilaterally.^{157,158} The futility debate ultimately condenses into a conflict in which patients or families and physicians disagree about the patient's right to receive a treatment that is highly unlikely to succeed. The futility concept fails to acknowledge the huge amount of excellent work done by ICU professionals to involve families in the decision-making process and to recognize the emotional and social problems raised by family participation in decision making.^{29,30} Rather than speak of the physician's right to withhold or withdraw treatment, we should speak of the physician's duty to prepare the ICU team and the family for a clearly identifiable shift from curative care to care aimed at optimizing patient comfort and dignity while alleviating distress in the family members.¹⁵⁹ ICU teams must move beyond the futility debate toward a position firmly rooted in a care, an ethic that gives a meaningful role to all staff members (physicians, nurses, and other health care professionals) in organizing the decision-making process with the patients and families, while interfacing with health care institutions, third-party payers, and, the general public.¹⁶⁰

Factors Associated with Decisions to Forego Life-Sustaining Treatment

As discussed in the previous section, the two main factors that should influence decisions to forgo life-sustaining treatments are patient preferences and the imminence of death (poor patient prognosis). An assessment of prognosis usually takes into account factors such as patient age, previous health status (cognitive function, self-sufficiency, and chronic morbidities), time spent in the ICU, severity at ICU admission, and worsening organ failure despite ICU management.^{7,9,11,73} Nevertheless, these factors may be overshadowed by events or perceptions that develop during an ICU stay,¹⁶¹ such as dependency on catecholamines and intensivists' perceptions that the prognosis is poor and the condition irreversible.¹⁰ Thus, multiple factors exist and may compete with one another, clearly indicating a need for a rigorous and collegial decision-making process based on consensus-building centered on the patient's and family's preferences.^{9,108,162,163} Several studies show that intensivists pay close attention to quality of life.^{7,9,74,92,94} Nevertheless, there is evidence that intensivists underestimate the quality of life of their patients, and that quality of life does not influence patient preferences regarding the intensity of treatment received in the ICU.¹⁶³

Studies have found that the frequency and nature of decisions to forego life-sustaining treatment vary with the personal characteristics of physicians and with their experience.^{28,164} As pointed out above, religious beliefs and cultural background play a role. In addition, physician gender,⁸⁹ specialty,¹⁶⁵ time working in ICUs,¹⁶⁶ working in private versus public institutions,¹⁶⁷ and teaching status of the hospital⁷⁸ influence practices regarding end-of-life care. It should be noted that participating in decisions to forego life-sustaining treatment and caring for dying patients has been associated with higher rates of burnout syndrome and conflict in the ICU.^{168,169} Conflict increases job strain,¹⁶⁸ and burnout syndrome has been associated with high employee turnover rates, absenteeism, and poor communication with families, all of which may impact the quality of a decision to forego life-sustaining treatment.¹⁶⁹

Christakis et al identified four major sources of bias affecting the nature of treatments withdrawn from patients in whom a decision to forego life-sustaining treatment had been made.¹⁷⁰ These biases may have clinical, social, and ethical consequences. Intensivists were more likely to withdraw treatments supporting organs that failed for chronic or "natural" reasons than for iatrogenic reasons, treatments started recently, and treatments whose withdrawal would result in immediate death, although when the diagnosis was uncertain, they preferred treatment withdrawals that resulted in delayed death. Similarly, Asch et al found that intensivists preferred to withdraw treatments that were expensive, scarce, and invasive.¹⁷¹ The incidence of decisions to forego life-sustaining treatment increases with ICU stay duration, indicating that intensivists respond rationally to persistent or worsening organ failures by adapting their management strategy.²⁰ Thus, Cook et al identified several time-dependent factors associated with withdrawal of mechanical ventilation: dependency on catecholamines, the physician's prediction of a low likelihood of surviving the ICU stay or a high likelihood of poor cognitive function, and physician perception that the patient did not want life support.¹⁰

Recent data on prolonged mechanical ventilation in ICU patients showed a high rate of major dependency among survivors.¹⁷²⁻¹⁷⁴ Asch et al reported that treatment withdrawal in the ICU usually occurs in the following order: blood products, dialysis, catecholamines, mechanical ventilation, parenteral nutrition, antibiotics, infusions, and enteral nutrition.¹⁷⁵ Whereas Cook et al found that mechanical ventilation was often stopped at the same time as dialysis or catecholamines,^{10,176} others reported that mechanical ventilation was sometimes continued after the withdrawal of dialysis and catecholamines.^{7,9} In France, nutrition and hydration are rarely withdrawn.^{7,9} Thus, even when a decision to forego life-sustaining treatment is made, the sequence of withdrawals is influenced by multiple, complex, nonclinical

factors that may seem irrelevant. These factors may be ethical, social, religious, or related to family preferences.⁹⁴ In one study, the same intensivists made contradictory decisions about the same clinical scenarios at different points in time.¹⁷⁷ Similarly, our group and others have shown that making a decision to forego life-sustaining treatment influences survival independently of all other factors known to affect survival, indicating that a decision to forego life-sustaining treatment depends in part on setting-specific characteristics that are difficult to identify.^{178–180}

WHO Decides?

End-of-life management of ICU patients can be artificially distinguished into two phases. The first phase is making the decision to forego life-sustaining treatment by building a consensus that treatment must move from curative interventions to palliative (comfort) care.²⁶ Ideally, this consensus is achieved using the shared decision-making model recommended by Society of Critical Care Medicine in 2001 and an international consensus conference in 2003.^{17,18} In this model, the decision to forego life-sustaining treatment is made by a well-organized team in which physicians and nurses communicate openly, provide relevant medical information, elicit patient values and preferences, explore the family's preferred role in the decision process, and achieve a consensus that is most consistent with the patient's values and preferences.¹⁸¹ From the beginning of the decision-making process, the family members participate, either by bearing witness to the patient's wishes or by explaining their own wishes.¹²

Although the current goal standard of decision making is a consensus shared by health care providers and surrogate decision makers acting in the patient's best interest, diverging opinions often emerge, and conflict may be inevitable.¹⁶⁸ A previous conflict, inadequate information, or economic restrictions imposed by managed care may lead families to view a decision to forego life-sustaining treatment with distrust.¹⁸² When such conflict occurs, physicians should refrain from unilaterally enforcing their own decision over the objections of the family.¹⁸³ On the contrary, physicians should intensify communication with families,³ initiate a process of negotiation,^{96,184–186} seek external advice, or show families that the decision is consistent with institutional policies and recommendations issued by learned societies.^{136,187} Physicians should must be aware that their own value judgments may be different than those of the patient, surrogates, and other members of the health care team such as nurses.^{188,189} Although surrogate decision makers fulfill a vital role in the decision-making process, as many as one-fourth of all deaths in the ICU occur in patients without decision-making capacity and without a surrogate decision maker. Few guidelines exist to guide a decision to forego life-sustaining treatment in such circumstances, and wide variability exists in how health care teams address a decision to forego life-sustaining treatment in this patient population, often without institutional oversight.¹⁹⁰

The second phase in the end-of-life management of ICU patients consists in implementing the decision to forego life-sustaining treatment. This requires an organized strategy relying on both human and technological resources to meet the expectations of patients and families. ICU teams must learn this approach, if needed with the help of palliative care teams or consulting ethicists, where available.^{17,18,20,46,71,162,184,187,191–195}

This phase may last a few hours to a few days. Support must be provided not only to families, but also to ICU staff members, who should be confident at all times that comfort care is preferable over the use of sophisticated technologies that prolong life artificially.¹⁹⁶ The ICU team may need to orchestrate the dying process by adjusting the treatment to the needs of the family (e.g., maintaining mechanical ventilation unchanged until the entire family is ready or a family member arrives from a far-away location).^{196,197}

Making and implementing a decision to forego life-sustaining treatment that is in the patient’s best interest, especially in the setting of complex negotiation, is a clinical procedure that must be learned, practiced, refined, and evaluated, just as are other ICU skills.¹⁹¹ Physicians must take into account both ethical principles and the dignity of patients and families.^{192,198,199} Because decisions to forego life-sustaining treatment are often irreversible and associated with imminent mortality, it is imperative that ICU physicians be trained and evaluated in assessing and implementing decisions to forego life-sustaining treatment, and in providing support to dying patients and their families.^{40,193,198,200–204}

The Intensive Care Unit Health Care Team: A Key Role in Communicating with Families of Dying Patients

Several studies have sought to identify the needs and expectations of families of ICU patients.^{35,36,39,41,205–207} The specific needs of families of dying patients have been investigated in studies of the overall long-term impact on families of the ICU experience and bereavement,^{27,43–45,208,209} in studies of interventions (the presence of families during resuscitation or involvement of families in decisions),^{30,210} and in studies of the impact of family conferences on the decision to forego life-sustaining treatment process.^{200,211} Families have stated that a pain-free death was a key priority,⁴⁹ and that they often perceived information as quantitatively or qualitatively inadequate. For instance, some families complained that they did not know the cause of death. These data have been used to identify specific needs of families of patients dying in the ICU (Tables 66-3 and 66-4).^{18,20} Keenan et al found that families had seven major needs directly related to the decision-making process: that the process be well explained; that it proceed as expected; that the patient be comfortable; that the discussion be initiated by a member of the ICU team; that the family and friends be prepared for the decision; that the family be allowed privacy during implementation of the decision to forego life-sustaining treatment;²¹² and that the family be given opportunities to voice concerns and to make special requests.²⁰⁸ Others have emphasized the importance of meeting the spiritual and religious needs of patients and family members.^{27,213,206,214}

Table 66-3: Ten Most Important Needs of Families of Critically Ill Dying Patients

- | |
|---|
| <ol style="list-style-type: none">1. To be with the person2. To be helpful to the dying person3. To be informed of the dying person’s changing condition4. To understand what is being done to the patient and why5. To be assured of the patient’s comfort6. To be comforted7. To express emotions8. To be assured that their decisions were right9. To find meaning in the dying of their loved one10. To be fed, hydrated, and rested |
|---|

Source: Used, with permission, from Truog et al.¹⁸

Table 66-4: A Dozen Needs of the Family in the Setting of Critical Illness

1. To have questions answered honestly
2. To know specific facts about what is wrong with the patient
3. To know the prognosis for recovery
4. To be called at home about changes in the patient's status
5. To receive information from the physician (at least) once daily
6. To receive information in understandable language
7. To believe that hospital personnel care about the patient
8. To be assured of the patient's comfort
9. To be comforted
10. To express emotions
11. To find meaning in the death of their loved one
12. To have the opportunity to eat, drink, and sleep

Source: Used, with permission, from Prendergast et al.²⁰

Meeting these needs empowers and motivates families to participate in the process of deciding to forego life-sustaining treatment. Burdens weighing on families should not be construed as reasons to exclude families from decision making.¹² Conversely, involving families at all costs without previously providing them with information and with psychological and social support may lead to severe residual disorders, including severe persistent guilt and abnormal bereavement.^{30,210} Thus, sharing decisions with families is intricately linked with a procedure of intensive communication that supports and guides the family at the time of the decision to forego life-sustaining treatment.^{46–48,187,195,215} The reasons justifying a decision to forego life-sustaining treatment and, subsequently, the specific implementation modalities are discussed, usually during family conferences attended by members of the family and members of the ICU team (physicians, nurses, and social workers).¹⁶² Informal conversations, however, at the bedside or in the hallway or waiting room are also useful and may involve a smaller number of people (often the intensivist and the patient's spouse). Nevertheless, formal family conferences are invaluable for strengthening communication between the family and the ICU team. They should be offered as often as needed, and at least once for each family.

During the conferences, the ICU team members are familiar faces for the family members, and the physician leading the conference starts by introducing each person. Each conference is prepared by the ICU team to establish a consensus among staff members^{201,216–219} as a prerequisite to the development of a consensus with the family.^{20,185,186,217} Regular decision-making and debriefing meetings of physicians and nurses are essential to ensure that all team members understand one another's points of view.¹² Similarly, because resident physicians receive insufficient training in ethical principles,²²⁰ informal discussions between residents and senior ICU physicians should be encouraged to improve residents' confidence in withdrawal decisions. Residents should also be encouraged to participate in family meetings.²²¹ During family conferences, the word "death" should be used and the manner in which the patient will die should be explained to the family members, with emphasis on patient comfort. The ICU staff must allow the family members to speak and to vent their emotions, but they must also anticipate questions the family members cannot bring themselves to ask.^{200,211} Studies show not only that inadequate information is a source of dissatisfaction among families of patients

dying in the ICU, but also that targeted interventions aimed at improving communication may be ineffective.^{50,215,222–228} In addition, clinicians lack training in techniques that help families consider decision sharing, express their opinions, and cope with the attendant burden.^{192,220,229,230}

A key objective of the family conference is to assure family members that the patient's wishes will be honored, although the patient is incompetent.¹⁹² Neither families nor health care professionals know how to predict patients' wishes regarding a decision to forego life-sustaining treatment in the ICU.^{139,141,147,219,231} In practice, the physicians must first ask the family whether the patient has expressed wishes about end-of-life care. If the patient has expressed wishes, then the family members describe them and use substituted judgment to participate in a decision to forego life-sustaining treatment. When nothing is known about the patient's wishes, the best interests of the patient guide the decision-making process. Placing the patient's wishes at the center of the discussions and negotiations ensures that decision making is of high moral and ethical quality despite the absence of patient participation.¹⁸⁶ Surrogate decision makers should receive active support from the ICU team.⁴⁶ Furthermore, by making it clear that families speak for their loved one and not for themselves, this approach should minimize feelings of guilt secondary to involvement in a decision to forego life-sustaining treatment.²³²

For most decisions to forego life-sustaining treatment, a consensus is achieved rapidly with the family.⁸² In difficult cases, a process of negotiation should be initiated to convince the family that the patient cannot survive and that the patient's best interests govern decisions.²³³ It may be possible to continue life support for an interval time so as to help the family understand that the patient's condition is irreversible. Compassion and understanding must sustain the relationship with families. Families are often dissatisfied with the level of communication, but are unable to express their dissatisfaction because they do not wish to provoke an open conflict. Intensive communication is the only road leading from a potential battle of diverging opinions to a consensus in which the family accepts, probably with pain and sadness, that the best care available for their loved one is discontinuation of aggressive artificial treatments and continuous painstaking attention to comfort. This awareness that treatments must be withheld or withdrawn is acquired via a multifaceted process whose components are cognitive (understanding the information and trusting the ICU team), emotional (believing that the patient is not being abandoned but has reached the end of his or her life),²⁰ and interpersonal (protecting the family and friends from long-term harm related to participation in decision making).^{209,234} At this stage, lifting restrictions on visiting hours and encouraging participation in patient care (washing, massaging, pain relief treatment, and so on) provides families with a sense of comfort, intimacy, and utility. Family members have a strong desire for opportunities to communicate with the patient (by touching, speaking, looking at photos, and listening to music). The health care professionals must respect the silence, rituals, emotions, and religious practices of each family. Again, everything must be done to minimize feelings of guilt.

Important Unknowns

Despite considerable advancements made in the past 20 years, significant barriers to optimal decision making regarding life support and end-of-life care in the ICU still exist. In a recent survey, nursing and physician directors of ICUs in the United States reported major barriers in the following areas: clinician training in effective communication with patients and patients' families, as well as with the medical team; competing demands for clinician time; unrealistic patient and family expectations about prognosis; inability of patients to participate in treatment decisions; and disagreements within families about goals of care.²³⁵ Despite a general consensus on basic principles that should guide the delivery of end-of-life care in the ICU, numerous studies show that the implementation of these principles is lagging. Many ICU physicians are

unaware of patient wishes,²³⁶ physicians do not document patient preferences,²³⁷ many patients still die in moderate or severe pain,⁵⁰ and half the families experience inadequate communication with physicians.³⁴ Physicians do not discuss basic prognosis for survival in one-third of all family conferences²³⁸ and lack basic communication skills, such as showing empathy²⁰⁷ and responding to family members' emotional cues.²³⁹ Additionally, many physicians are reluctant to give a recommendation regarding decisions to forego life sustaining treatment, even when directly asked to do so by surrogates.²⁴⁰ Families with limited English proficiency may be at risk for further communication deficiencies.²⁴¹ Various solutions to the aforementioned barriers have been proposed,²³⁵ and these solutions form the emphasis of current investigations.

The Future

In the future, intensive multidisciplinary programs and early incorporation of palliative care into ICU management should be encouraged so as to improve information and communication. Multidisciplinary teams could either handle communication with families of dying patients or assist the ICU staff members and teach them palliative care strategies.

Data from the literature strongly support a heavy emphasis on palliative care.^{242,243} By facilitating prognostic information and promoting communication with patients and families, huge efforts on the part of investigators have been successful in identifying patient preferences and wishes regarding treatment intensity. A program of intensive palliative care offered to dying ICU patients holds promise as a means of improving communication at a time when openness is crucial.^{47,48,187,242,243} Nevertheless, the generalizability and feasibility of this strategy require evaluation in a range of countries and cultures.⁵⁰

Summary and Conclusion

Mechanical ventilation is the life-sustaining treatment most often used in the ICU. Most patients are successfully weaned off the ventilator. A few, however, cannot recover from their disease, remain dependent on the ventilator, and have a very high likelihood of death in the short term. In this situation, intensivists must engage in the complex process of deciding to withdraw mechanical ventilation. These patients must be offered the best possible death. We must remain at their side and on their side, keeping them comfortable at all times, while encouraging family involvement in decisions and care, if they so wish, welcoming family members in the ICU as partners rather than as visitors, alleviating their guilt, and helping them to prepare for the bereavement process.

Over the last 20 years, several descriptive studies have provided detailed epidemiologic data on practices and changes to practices over time, as well as on targets for improvement. Qualitative studies have helped us to understand the complexity of the decision-making process, to recognize that upholding ethical principles requires more than compliance with the law, and to distinguish intentions from practices. The results of these studies indicate a need for major improvements in end-of-life care for ICU patients, most notably regarding our skill in communicating, imparting information, and organizing and implementing treatment withholding and withdrawal. Further knowledge of the decision-making process and of the individuals involved in it will help to improve our end-of-life practices, rest our decisions to forego life-sustaining treatment on ethical principles, and validate the shared decision-making model by studies of outcomes in family members long after the ICU experience. Serving the principle of autonomy requires not only that a patient's wishes be honored, but also that the family be able to proceed through the bereavement process with the least possible guilt, remorse, and regret. Thus, our end-of-life practice should be tailored to the specific characteristics of each region and culture, each clinical situation, each patient, each family, and each ICU professional.

References

1. Ruark JE, Raffin TA. Initiating and withdrawing life support. Principles and practice in adult medicine. *N Engl J Med*. 1988;318:25–30. [[PubMed: 3275889](#)]

2. Emanuel EJ. Should physicians withhold life-sustaining care from patients who are not terminally ill? *Lancet*. 1988;1:106–108. [[PubMed: 2891943](#)]

3. Burns JP, Edwards J, Johnson J, et al. Do-not-resuscitate order after 25 years. *Crit Care Med*. 2003;31:1543–1550. [[PubMed: 12771631](#)]

4. Tobin MJ. Advances in mechanical ventilation. *N Engl J Med*. 2001;344:1986–1996. [[PubMed: 11430329](#)]

5. Jayes RL, Zimmerman JE, Wagner DP, et al. Do-not-resuscitate orders in intensive care units. Current practices and recent changes. *JAMA*. 1993;270:2213–2217. [[PubMed: 8411606](#)]

6. Hall RI, Rocker GM. End-of-life care in the ICU: treatments provided when life support was or was not withdrawn. *Chest*. 2000;118:1424–1430. [[PubMed: 11083696](#)]

7. Ferrand E, Robert R, Ingrand P, Lemaire F. Withholding and withdrawal of life support in intensive-care units in France: a prospective survey. French LATAREA Group. *Lancet*. 2001;357:9–14. [[PubMed: 11197395](#)]

8. Lee DK, Swinburne AJ, Fedullo AJ, Wahl GW. Withdrawing care. Experience in a medical intensive care unit. *JAMA*. 1994;271:1358–1361. [[PubMed: 8158822](#)]

9. Pochard F, Azoulay E, Chevret S, et al. French intensivists do not apply American recommendations regarding decisions to forgo life-sustaining therapy. *Crit Care Med*. 2001;29:1887–1892. [[PubMed: 11588446](#)]

10. Cook D, Rocker G, Marshall J, et al. Withdrawal of mechanical ventilation in anticipation of death in the intensive care unit. *N Engl J Med*. 2003;349:1123–1132. [[PubMed: 13679526](#)]

11. Smedira NG, Evans BH, Grais LS, et al. Withholding and withdrawal of life support from the critically ill. *N Engl J Med*. 1990;322:309–315. [[PubMed: 2296273](#)]

12. Azoulay E, Sprung CL. Family-physician interactions in the intensive care unit. *Crit Care Med*. 2004;32:2323–2328. [[PubMed: 15640649](#)]

13. Cook D. Patient autonomy versus parentalism. *Crit Care Med*. 2001; 29:N24–N25.

14. Luce JM. New standards for patient rights and medical competence. *Crit Care Med*. 2000;28:3114–3115. [[PubMed: 10966322](#)]

15. Way J, Back AL, Curtis JR. Withdrawing life support and resolution of conflict with families. *BMJ*. 2002;325:1342–1345. [[PubMed: 12468486](#)]

16. Beauchamp TL. Methods and principles in biomedical ethics. *J Med Ethics*. 2003;29:269–274. [[PubMed: 14519835](#)]

17. Carlet J, Thijs LG, Antonelli M, et al. Challenges in end-of-life care in the ICU. Statement of the 5th International Consensus Conference in Critical Care: Brussels, Belgium, April 2003. *Intensive Care Med.* 2004;30:770–784. [[PubMed: 15098087](#)]

18. Truog RD, Cist AF, Brackett SE, et al. Recommendations for end-of-life care in the intensive care unit: the Ethics Committee of the Society of Critical Care Medicine. *Crit Care Med.* 2001;29:2332–2348. [[PubMed: 11801837](#)]

19. Luce JM, Alpers A. End-of-life care: what do the American courts say? *Crit Care Med.* 2001;29:N40–N45.

20. Prendergast TJ, Puntillo KA. Withdrawal of life support: intensive caring at the end of life. *JAMA.* 2002;288:2732–2740. [[PubMed: 12460097](#)]

21. Singer PA, Lowy FH. Rationing, patient preferences, and cost of care at the end of life. *Arch Intern Med.* 1992;152:478–480. [[PubMed: 1546909](#)]

22. Good care of the dying patient. Council on Scientific Affairs, American Medical Association. *JAMA.* 1996;275:474–478.

23. Quill TE. Perspectives on care at the close of life. Initiating end-of-life discussions with seriously ill patients: addressing the “elephant in the room.” *JAMA.* 2000;284:2502–2507. [[PubMed: 11074781](#)]

24. Emanuel LL. Facing requests for physician-assisted suicide: toward a practical and principled clinical skill set. *JAMA.* 1998;280:643–647. [[PubMed: 9718058](#)]

25. Emanuel EJ, Daniels ER, Fairclough DL, Clarridge BR. The practice of euthanasia and physician-assisted suicide in the United States: adherence to proposed safeguards and effects on physicians. *JAMA.* 1998;280:507–513. [[PubMed: 9707132](#)]

26. Faber-Langendoen K, Lanken PN. Dying patients in the intensive care unit: forgoing treatment, maintaining care. *Ann Intern Med.* 2000;133:886–893. [[PubMed: 11103059](#)]

27. Abbott KH, Sago JG, Breen CM, et al. Families looking back: one year after discussion of withdrawal or withholding of life-sustaining support. *Crit Care Med.* 2001;29:197–201. [[PubMed: 11176185](#)]

28. Quill TE, Brody H. Physician recommendations and patient autonomy: finding a balance between physician power and patient choice. *Ann Intern Med.* 1996;125:763–769. [[PubMed: 8929011](#)]

29. Azoulay E, Pochard F, Chevret S, et al. Half the family members of intensive care unit patients do not want to share in the decision-making process: a study in 78 French intensive care units. *Crit Care Med.* 2004;32:1832–1838. [[PubMed: 15343009](#)]

30. Azoulay E, Pochard F, Kentish-Barnes N, et al. Risk of post-traumatic stress symptoms in family members of intensive care unit patients. *Am J Respir Crit Care Med.* 2005;171:987–994. [[PubMed: 15665319](#)]

31. Sprung CL, Eidelman LA. Worldwide similarities and differences in the foregoing of life-sustaining treatments. *Intensive Care Med.* 1996;22:1003–1005. [[PubMed: 8923060](#)]

32. Campbell ML, Bizek KS, Thill M. Patient responses during rapid terminal weaning from mechanical ventilation: a prospective study. *Crit Care Med.* 1999;27:73–77. [[PubMed: 9934896](#)]

-
33. Tonelli MR. Waking the dying: must we always attempt to involve critically ill patients in end-of-life decisions? *Chest*. 2005;127:637–642. [[PubMed: 15706007](#)]
-
34. Azoulay E, Chevret S, Leleu G, et al. Half the families of intensive care unit patients experience inadequate communication with physicians. *Crit Care Med*. 2000;28:3044–3049. [[PubMed: 10966293](#)]
-
35. Johnson D, Wilson M, Cavanaugh B, et al. Measuring the ability to meet family needs in an intensive care unit. *Crit Care Med*. 1998;26:266–271. [[PubMed: 9468163](#)]
-
36. Azoulay E, Pochard F, Chevret S, et al. Meeting the needs of intensive care unit patient families: a multicenter study. *Am J Respir Crit Care Med*. 2001;163:135–139. [[PubMed: 11208638](#)]
-
37. Pochard F, Azoulay E, Chevret S, et al. Symptoms of anxiety and depression in family members of intensive care unit patients: ethical hypothesis regarding decision-making capacity. *Crit Care Med*. 2001;29:1893–1897. [[PubMed: 11588447](#)]
-
38. Epstein RM, Alper BS, Quill TE. Communicating evidence for participatory decision making. *JAMA*. 2004;291:2359–2366. [[PubMed: 15150208](#)]
-
39. Heyland DK, Tranmer JE. Measuring family satisfaction with care in the intensive care unit: the development of a questionnaire and preliminary results. *J Crit Care*. 2001;16:142–149. [[PubMed: 11815899](#)]
-
40. Sjobqvist P, Nilstun T, Svantesson M, Berggren L. Withdrawal of life support—who should decide? Differences in attitudes among the general public, nurses and physicians. *Intensive Care Med*. 1999;25:949–954. [[PubMed: 10501750](#)]
-
41. Wasser T, Pasquale MA, Matchett SC, et al. Establishing reliability and validity of the critical care family satisfaction survey. *Crit Care Med*. 2001;29:192–196. [[PubMed: 11176184](#)]
-
42. Hanson LC, Danis M, Garrett JM, Mutran E. Who decides? Physicians' willingness to use life-sustaining treatment. *Arch Intern Med*. 1996;156:785–789. [[PubMed: 8615712](#)]
-
43. Cuthbertson SJ, Margetts MA, Streat SJ. Bereavement follow-up after critical illness. *Crit Care Med*. 2000;28:1196–1201. [[PubMed: 10809305](#)]
-
44. Covinsky KE, Goldman L, Cook EF, et al. The impact of serious illness on patients' families. SUPPORT Investigators. Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatment. *JAMA*. 1994;272:1839–1844. [[PubMed: 7990218](#)]
-
45. Malacrida R, Bettelini CM, Degrate A, et al. Reasons for dissatisfaction: a survey of relatives of intensive care patients who died. *Crit Care Med*. 1998;26:1187–1193. [[PubMed: 9671367](#)]
-
46. Lilly CM, De Meo DL, Sonna LA, et al. An intensive communication intervention for the critically ill. *Am J Med*. 2000;109:469–475. [[PubMed: 11042236](#)]
-
47. Dowdy MD, Robertson C, Bander JA. A study of proactive ethics consultation for critically and terminally ill patients with extended lengths of stay. *Crit Care Med*. 1998;26:252–259. [[PubMed: 9468161](#)]
-

48. Schneiderman LJ, Gilmer T, Teetzel HD. Impact of ethics consultations in the intensive care setting: a randomized, controlled trial. *Crit Care Med*. 2000;28:3920–3924. [[PubMed: 11153636](#)]

49. Lynn J, Teno JM, Phillips RS, et al. Perceptions by family members of the dying experience of older and seriously ill patients. SUPPORT Investigators. Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments. *Ann Intern Med*. 1997;126:97–106. [[PubMed: 9005760](#)]

50. A controlled trial to improve care for seriously ill hospitalized patients. The study to understand prognoses and preferences for outcomes and risks of treatments (SUPPORT). The SUPPORT Principal Investigators. *JAMA*. 1995;274:1591–1598.

51. Quill TE, Dresser R, Brock DW. The rule of double effect—a critique of its role in end-of-life decision making. *N Engl J Med*. 1997;337:1768–1771. [[PubMed: 9392707](#)]

52. Luce JM, Alpers A. Legal aspects of withholding and withdrawing life support from critically ill patients in the United States and providing palliative care to them. *Am J Respir Crit Care Med*. 2000;162:2029–2032. [[PubMed: 11112108](#)]

53. Sulmasy DP, Pellegrino ED. The rule of double effect: clearing up the double talk. *Arch Intern Med*. 1999;159:545–550. [[PubMed: 10090110](#)]

54. Sulmasy DP, Sood JR, Ury WA. The quality of care plans for patients with do-not-resuscitate orders. *Arch Intern Med*. 2004;164:1573–1578. [[PubMed: 15277292](#)]

55. Quill TE. The ambiguity of clinical intentions. *N Engl J Med*. 1993;329:1039–1040. [[PubMed: 7690111](#)]

56. Keenan SP, Busche KD, Chen LM, et al. A retrospective review of a large cohort of patients undergoing the process of withholding or withdrawal of life support. *Crit Care Med*. 1997;25:1324–1331. [[PubMed: 9267945](#)]

57. Quill TE. Barbiturates in the care of the terminally ill. *N Engl J Med*. 1993;328:1350; author reply 1351. [[PubMed: 7682289](#)]

58. Holzapfel L, Demingon G, Piralla B, et al. A four-step protocol for limitation of treatment in terminal care. An observational study in 475 intensive care unit patients. *Intensive Care Med*. 2002;28:1309–1315. [[PubMed: 12209282](#)]

59. Brody H, Campbell ML, Faber-Langendoen K, Ogle KS. Withdrawing intensive life-sustaining treatment—recommendations for compassionate clinical management. *N Engl J Med*. 1997;336:652–657. [[PubMed: 9032053](#)]

60. Krishna G, Raffin TA. Terminal weaning from mechanical ventilation. *Crit Care Med*. 1999;27:9–10. [[PubMed: 9934874](#)]

61. Faber-Langendoen K. The clinical management of dying patients receiving mechanical ventilation. A survey of physician practice. *Chest*. 1994;106:880–888. [[PubMed: 8082372](#)]

62. Cook DJ, Meade MO, Perry AG. Qualitative studies on the patient's experience of weaning from mechanical ventilation. *Chest*. 2001; 120:469S–473S.

63. Barnato AE, Herndon MB, Anthony DL, et al. Are regional variations in end-of-life care intensity explained by patient preferences? A study of the US Medicare population. *Med Care*. 2007;45:386–393. [[PubMed: 17446824](#)]

-
64. Pronovost P, Angus DC. Economics of end-of-life care in the intensive care unit. *Crit Care Med*. 2001;29:N46–N51.
-
65. Guidelines for intensive care unit admission, discharge, and triage. Task Force of the American College of Critical Care Medicine, Society of Critical Care Medicine. *Crit Care Med*. 1999;27:633–638.
-
66. Understanding costs and cost-effectiveness in critical care. Report from the second American thoracic society workshop on outcomes research. *Am J Respir Crit Care Med*. 2002;165:540–550.
-
67. Luce JM, Rubenfeld GD. Can health care costs be reduced by limiting intensive care at the end of life? *Am J Respir Crit Care Med*. 2002; 165:750–754. [[PubMed: 11897638](#)]
-
68. Danis M, Mutran E, Garrett JM, et al. A prospective study of the impact of patient preferences on life-sustaining treatment and hospital cost. *Crit Care Med*. 1996;24:1811–1817. [[PubMed: 8917030](#)]
-
69. Emanuel EJ. Cost savings at the end of life. What do the data show? *JAMA*. 1996;275:1907–1914. [[PubMed: 8648872](#)]
-
70. Studdert DM, Mello MM, Burns JP, et al. Conflict in the care of patients with prolonged stay in the ICU: types, sources, and predictors. *Intensive Care Med*. 2003;29:1489–1497. [[PubMed: 12879243](#)]
-
71. Burns JP, Mello MM, Studdert DM, et al. Results of a clinical trial on care improvement for the critically ill. *Crit Care Med*. 2003;31:2107–2117. [[PubMed: 12973167](#)]
-
72. Zhang B, Wright AA, Huskamp HA, et al. Health care costs in the last week of life: associations with end-of-life conversations. *Arch Intern Med*. 2009;169:480–488. [[PubMed: 19273778](#)]
-
73. Sprung CL, Cohen SL, Sjøkvist P, et al. End-of-life practices in European intensive care units: the Ethicus Study. *JAMA*. 2003;290:790–797. [[PubMed: 12915432](#)]
-
74. Vincent JL. Forgoing life support in western European intensive care units: the results of an ethical questionnaire. *Crit Care Med*. 1999;27:1626–1633. [[PubMed: 10470775](#)]
-
75. Prendergast TJ, Claessens MT, Luce JM. A national survey of end-of-life care for critically ill patients. *Am J Respir Crit Care Med*. 1998; 158:1163–1167. [[PubMed: 9769276](#)]
-
76. Esteban A, Gordo F, Solsona JF, et al. Withdrawing and withholding life support in the intensive care unit: a Spanish prospective multi-centre observational study. *Intensive Care Med*. 2001;27:1744–1749. [[PubMed: 11810117](#)]
-
77. Eidelman LA, Jakobson DJ, Worner TM, et al. End-of-life intensive care unit decisions, communication, and documentation: an evaluation of physician training. *J Crit Care*. 2003;18:11–16. [[PubMed: 12640607](#)]
-
78. Keenan SP, Busche KD, Chen LM, et al. Withdrawal and withholding of life support in the intensive care unit: a comparison of teaching and community hospitals. The Southwestern Ontario Critical Care Research Network. *Crit Care Med*. 1998;26:245–251. [[PubMed: 9468160](#)]
-
79. Faber-Langendoen K, Bartels DM. Process of forgoing life-sustaining treatment in a university hospital: an empirical study. *Crit Care Med*. 1992;20:570–577. [[PubMed: 1374002](#)]
-

80. Faber-Langendoen K. A multi-institutional study of care given to patients dying in hospitals. Ethical and practice implications. *Arch Intern Med*. 1996;156:2130–2136. [[PubMed: 8862106](#)]
-
81. Meissner A, Genga KR, Studart FS, et al. Epidemiology of and factors associated with end-of-life decisions in a surgical intensive care unit. *Crit Care Med*. 2010;38:1060–1068. [[PubMed: 20068464](#)]
-
82. Prendergast TJ, Luce JM. Increasing incidence of withholding and withdrawal of life support from the critically ill. *Am J Respir Crit Care Med*. 1997;155:15–20. [[PubMed: 9001282](#)]
-
83. Jakobson DJ, Eidelman LA, Worner TM, et al. Evaluation of changes in forgoing life-sustaining treatment in Israeli ICU patients. *Chest*. 2004;126:1969–1973. [[PubMed: 15596700](#)]
-
84. Azoulay E, Adrie C, De Lassence A, et al. Determinants of postintensive care unit mortality: a prospective multicenter study. *Crit Care Med*. 2003;31:428–432. [[PubMed: 12576947](#)]
-
85. Cooke CR, Hotchkin DL, Engelberg RA, et al. Predictors of time to death after terminal withdrawal of mechanical ventilation in the ICU. *Chest*. 2010;138:289–297. [[PubMed: 20363840](#)]
-
86. Withholding and withdrawing life-sustaining therapy. This official statement of the American Thoracic Society was adopted by the ATS board of directors, March 1991. *Am Rev Respir Dis*. 1991;144:726–731.
-
87. Curtis JR, Vincent JL. Ethics and end-of-life care for adults in the intensive care unit. *Lancet*. 2010;376:1347–1353. [[PubMed: 20934213](#)]
-
88. Asai A, Fukuhara S, Lo B. Attitudes of Japanese and Japanese-American physicians towards life-sustaining treatment. *Lancet*. 1995;346:356–359. [[PubMed: 7623536](#)]
-
89. Mebane EW, Oman RF, Kroonen LT, Goldstein MK. The influence of physician race, age, and gender on physician attitudes toward advance care directives and preferences for end-of-life decision-making. *J Am Geriatr Soc*. 1999;47:579–591. [[PubMed: 10323652](#)]
-
90. Vincent JL. Cultural differences in end-of-life care. *Crit Care Med*. 2001;29:N52–N55.
-
91. Abizanda R, Almendros Corral L, Balerdi Perez B. Ethical aspects of intensive medicine. Results of an opinion survey [in Spanish]. *Med Clin (Barc)*. 1994;102:521–526. [[PubMed: 8208020](#)]
-
92. Asch DA, Hansen-Flaschen J, Lanken PN. Decisions to limit or continue life-sustaining treatment by critical care physicians in the United States: conflicts between physicians' practices and patients' wishes. *Am J Respir Crit Care Med*. 1995;151:288–292. [[PubMed: 7842181](#)]
-
93. Asch DA, DeKay ML. Euthanasia among US critical care nurses. Practices, attitudes, and social and professional correlates. *Med Care*. 1997;35:890–900. [[PubMed: 9298078](#)]
-
94. Cook DJ, Guyatt GH, Jaeschke R, et al. Determinants in Canadian health care workers of the decision to withdraw life support from the critically ill. Canadian Critical Care Trials Group. *JAMA*. 1995;273:703–708. [[PubMed: 7853627](#)]
-

95. Heyland DK, Rocker GM, O'Callaghan CJ, et al. Dying in the ICU: perspectives of family members. *Chest*. 2003;124:392–397. [[PubMed: 12853551](#)]
-
96. Teres D. Trends from the United States with end of life decisions in the intensive care unit. *Intensive Care Med*. 1993;19:316–322. [[PubMed: 8227721](#)]
-
97. Wenger NS, Pearson ML, Desmond KA, et al. Epidemiology of do-not-resuscitate orders. Disparity by age, diagnosis, gender, race, and functional impairment. *Arch Intern Med*. 1995;155:2056–2062. [[PubMed: 7575064](#)]
-
98. Giannini A, Pessina A, Tacchi EM. End-of-life decisions in intensive care units: attitudes of physicians in an Italian urban setting. *Intensive Care Med*. 2003;29:1902–1910 Epub 2003 Sep 11. [[PubMed: 13680120](#)]
-
99. Cardoso T, Fonseca T, Pereira S, Lencastre L. Life-sustaining treatment decisions in Portuguese intensive care units: a national survey of intensive care physicians. *Crit Care*. 2003;7:R167–R175.
-
100. Nava S, Sturani C, Hartl S, et al. End-of-life decision-making in respiratory intermediate care units: a European survey. *Eur Respir J*. 2007;30:156–164. [[PubMed: 17601972](#)]
-
101. Bloomer MJ, Tiruvoipati R, Tsiripillis M, Botha JA. End of life management of adult patients in an Australian metropolitan intensive care unit: a retrospective observational study. *Aust Crit Care*. 2010;23:13–19. [[PubMed: 19914844](#)]
-
102. Wunsch H, Linde-Zwirble WT, Harrison DA, et al. Use of intensive care services during terminal hospitalizations in England and the United States. *Am J Respir Crit Care Med*. 2009;180:875–880. [[PubMed: 19713448](#)]
-
103. Mani RK, Mandal AK, Bal S, et al. End-of-life decisions in an Indian intensive care unit. *Intensive Care Med*. 2009;35:1713–1719. [[PubMed: 19568731](#)]
-
104. Bertolini G, Boffelli S, Malacarne P, et al. End-of-life decision-making and quality of ICU performance: an observational study in 84 Italian units. *Intensive Care Med*. 2010;36:1495–1504. [[PubMed: 20464541](#)]
-
105. Brieva JL, Cooray P, Rowley M. Withholding and withdrawal of life-sustaining therapies in intensive care: an Australian experience. *Crit Care Resusc*. 2009;11:266–268. [[PubMed: 20001875](#)]
-
106. van der Maas PJ, van der Wal G, Haverkate I, et al. Euthanasia, physician-assisted suicide, and other medical practices involving the end of life in the Netherlands, 1990–1995. *N Engl J Med*. 1996;335:1699–1705.
-
107. Consensus report on the ethics of foregoing life-sustaining treatments in the critically ill. Task Force on Ethics of the Society of Critical Care Medicine. *Crit Care Med*. 1990;18:1435–1439.
-
108. Vincent JL. Information in the ICU: are we being honest with our patients? The results of a European questionnaire. *Intensive Care Med*. 1998;24:1251–1256. [[PubMed: 9885876](#)]
-
109. Emanuel EJ, Emanuel LL. Four models of the physician-patient relationship. *JAMA*. 1992;267:2221–2226. [[PubMed: 1556799](#)]
-
110. Seale C. Hastening death in end-of-life care: a survey of doctors. *Soc Sci Med*. 2009;69:1659–1666. [[PubMed: 19837498](#)]

-
111. Loggers ET, Maciejewski PK, Paulk E, et al. Racial differences in predictors of intensive end-of-life care in patients with advanced cancer. *J Clin Oncol*. 2009;27:5559–5564. [[PubMed: 19805675](#)]
-
112. Sprung CL, Maia P, Bulow HH, et al. The importance of religious affiliation and culture on end-of-life decisions in European intensive care units. *Intensive Care Med*. 2007;33:1732–1739. [[PubMed: 17541550](#)]
-
113. McLean RF, Tarshis J, Mazer CD, Szalai JP. Death in two Canadian intensive care units: institutional difference and changes over time. *Crit Care Med*. 2000;28:100–103. [[PubMed: 10667506](#)]
-
114. Angus DC, Barnato AE, Linde-Zwirble WT, et al. Use of intensive care at the end of life in the United States: an epidemiologic study. *Crit Care Med*. 2004;32:638–643. [[PubMed: 15090940](#)]
-
115. Sprung CL, Geber D, Eidelman LA, et al. Evaluation of triage decisions for intensive care admission. *Crit Care Med*. 1999;27:1073–1079. [[PubMed: 10397207](#)]
-
116. Azoulay E, Pochard F, Chevret S, et al. Compliance with triage to intensive care recommendations. *Crit Care Med*. 2001;29:2132–2136. [[PubMed: 11700409](#)]
-
117. Garrouste-Orgeas M, Montuclard L, Timsit JF, et al. Triaging patients to the ICU: a pilot study of factors influencing admission decisions and patient outcomes. *Intensive Care Med*. 2003;29:774–781. [[PubMed: 12677368](#)]
-
118. Wu SC, Chen JS, Wang HM, et al. Determinants of ICU care in the last month of life for Taiwanese cancer decedents, 2001 to 2006. *Chest*. 2010;138:1071–1077. [[PubMed: 20363837](#)]
-
119. Sharma G, Freeman J, Zhang D, Goodwin JS. Continuity of care and intensive care unit use at the end of life. *Arch Intern Med*. 2009;169:81–86. [[PubMed: 19139328](#)]
-
120. Seferian EG, Afessa B. Adult intensive care unit use at the end of life: a population-based study. *Mayo Clin Proc*. 2006;81:896–901. [[PubMed: 16835969](#)]
-
121. Teno JM, Clarridge BR, Casey V, et al. Family perspectives on end-of-life care at the last place of care. *JAMA*. 2004;291:88–93. [[PubMed: 14709580](#)]
-
122. Wright AA, Keating NL, Balboni TA, et al. Place of death: correlations with quality of life of patients with cancer and predictors of bereaved caregivers' mental health. *J Clin Oncol*. 2010;28:4457–4464. [[PubMed: 20837950](#)]
-
123. Barnato AE, Chang CC, Farrell MH, et al. Is survival better at hospitals with higher “end-of-life” treatment intensity? *Med Care*. 2010;48:125–132. [[PubMed: 20057328](#)]
-
124. Barnato AE, Berhane Z, Weissfeld LA, et al. Racial variation in end-of-life intensive care use: a race or hospital effect? *Health Serv Res*. 2006;41:2219–2237. [[PubMed: 17116117](#)]
-
125. Hanchate A, Kronman AC, Young-Xu Y, et al. Racial and ethnic differences in end-of-life costs: why do minorities cost more than whites? *Arch Intern Med*. 2009;169:493–501. [[PubMed: 19273780](#)]
-
126. Buckley TA, Joynt GM. Limitation of life support in the critically ill: the Hong Kong perspective. *Ann Acad Med Singapore*. 2001;30:281–286. [[PubMed: 11455743](#)]

-
127. The Belgian act on euthanasia of May, 28th 2002. *Eur J Health Law*. 2003;10:329–335.
-
128. Nys H. A presentation of the Belgian act on euthanasia against the background of Dutch euthanasia law. *Eur J Health Law*. 2003;10:239–255. [[PubMed: 14733042](#)]
-
129. Damas F, Damas P, Lamy M. Euthanasia: a law in Belgium? *Intensive Care Med*. 2001;27:1683. [[PubMed: 11685317](#)]
-
130. Lemaire FJ. A law for end of life care in France? *Intensive Care Med*. 2004;30:2120. [[PubMed: 15448892](#)]
-
131. Luce JM. Three patients who asked that life support be withheld or withdrawn in the surgical intensive care unit. *Crit Care Med*. 2002;30:775–780. [[PubMed: 11940744](#)]
-
132. Meier DE, Morrison RS. Autonomy reconsidered. *N Engl J Med*. 2002; 346:1087–1089. [[PubMed: 11932479](#)]
-
133. Alpers A, Lo B. Does it make clinical sense to equate terminally ill patients who require life-sustaining interventions with those who do not? *JAMA*. 1997;277:1705–1708. [[PubMed: 9169898](#)]
-
134. Whitney SN, McGuire AL, McCullough LB. A typology of shared decision making, informed consent, and simple consent. *Ann Intern Med*. 2004;140:54–59. [[PubMed: 14706973](#)]
-
135. Kirchhoff KT, Anumandla PR, Foth KT, et al. Documentation on withdrawal of life support in adult patients in the intensive care unit. *Am J Crit Care*. 2004;13:328–334. [[PubMed: 15293586](#)]
-
136. Luce JM. Making decisions about the forgoing of life-sustaining therapy. *Am J Respir Crit Care Med*. 1997;156:1715–1718. [[PubMed: 9412545](#)]
-
137. Luce JM, Lemaire F. Two transatlantic viewpoints on an ethical quandary. *Am J Respir Crit Care Med*. 2001;163:818–821. [[PubMed: 11282750](#)]
-
138. Heyland DK, Cook DJ, Rocker GM, et al. Decision-making in the ICU: perspectives of the substitute decision-maker. *Intensive Care Med*. 2003;29:75–82. [[PubMed: 12528026](#)]
-
139. Upadya A, Muralidharan V, Thorevska N, et al. Patient, physician, and family member understanding of living wills. *Am J Respir Crit Care Med*. 2002;166:1430–1435. [[PubMed: 12406822](#)]
-
140. Rabow MW, Hauser JM, Adams J. Supporting family caregivers at the end of life: “they don’t know what they don’t know.” *JAMA*. 2004;291:483–491. [[PubMed: 14747506](#)]
-
141. Hines SC, Glover JJ, Babrow AS, et al. Improving advance care planning by accommodating family preferences. *J Palliat Med*. 2001;4:481–489. [[PubMed: 11798480](#)]
-
142. Terry PB, Vettese M, Song J, et al. End-of-life decision making: when patients and surrogates disagree. *J Clin Ethics*. 1999;10:286–293. [[PubMed: 10791278](#)]
-
143. Emanuel EJ, Emanuel LL, Orentlicher D. Advance directives. *JAMA*. 1991;266:2563. [[PubMed: 1942400](#)]
-
144. Prendergast TJ. Advance care planning: pitfalls, progress, promise. *Crit Care Med*. 2001;29:N34–N399.

-
145. Danis M, Southerland LI, Garrett JM, et al. A prospective study of advance directives for life-sustaining care. *N Engl J Med*. 1991;324:882–888. [[PubMed: 2000110](#)]
-
146. Goodman MD, Tarnoff M, Slotman GJ. Effect of advance directives on the management of elderly critically ill patients. *Crit Care Med*. 1998;26:701–704. [[PubMed: 9559607](#)]
-
147. Sulmasy DP, Terry PB, Weisman CS, et al. The accuracy of substituted judgments in patients with terminal diagnoses. *Ann Intern Med*. 1998;128:621–629. [[PubMed: 9537935](#)]
-
148. Knaus WA, Zimmerman JE, Wagner DP, et al. APACHE-acute physiology and chronic health evaluation: a physiologically based classification system. *Crit Care Med*. 1981;9:591–597. [[PubMed: 7261642](#)]
-
149. Knaus WA, Draper EA, Wagner DP, Zimmerman JE. APACHE II: a severity of disease classification system. *Crit Care Med*. 1985;13:818–829. [[PubMed: 3928249](#)]
-
150. Le Gall JR, Lemeshow S, Saulnier F. A new Simplified Acute Physiology Score (SAPS II) based on a European/North American multicenter study. *JAMA*. 1993;270:2957–2963.
-
151. Le Gall JR, Klar J, Lemeshow S, et al. The Logistic Organ Dysfunction system. A new way to assess organ dysfunction in the intensive care unit. ICU Scoring Group. *JAMA*. 1996;276:802–810.
-
152. Timsit JF, Fosse JP, Troche G, et al. Accuracy of a composite score using daily SAPS II and LOD scores for predicting hospital mortality in ICU patients hospitalized for more than 72 h. *Intensive Care Med*. 2001;27:1012–1021. [[PubMed: 11497133](#)]
-
153. Timsit JF, Fosse JP, Toche G, et al. Calibration and discrimination by daily Logistic Organ Dysfunction scoring comparatively with daily Sequential Organ Failure Assessment scoring for predicting hospital mortality in critically ill patients. *Crit Care Med*. 2002;20:2003–2013.
-
154. Sculier JP, Paesmans M, Markiewicz E, Berghmans T. Scoring systems in cancer patients admitted for an acute complication in a medical intensive care unit. *Crit Care Med*. 2000;28:2786–2792. [[PubMed: 10966251](#)]
-
155. Schneiderman LJ, Spragg RG. Ethical decisions in discontinuing mechanical ventilation. *N Engl J Med*. 1988;318:984–988. [[PubMed: 3352689](#)]
-
156. Schneiderman LJ, Jecker NS, Jonsen AR. Medical futility: its meaning and ethical implications. *Ann Intern Med*. 1990;112:949–954. [[PubMed: 2187394](#)]
-
157. Schneiderman LJ, Jecker NS, Jonsen AR. Medical futility: response to critiques. *Ann Intern Med*. 1996;125:669–674. [[PubMed: 8849152](#)]
-
158. Prendergast TJ. Futility and the common cold. How requests for antibiotics can illuminate care at the end of life. *Chest*. 1995;107:836–844. [[PubMed: 7874961](#)]
-
159. Schneiderman LJ, Jecker NS, Jonsen AR. Abuse of futility. *Arch Intern Med*. 2001;161:128–130. [[PubMed: 11146716](#)]
-

160. Schneiderman LJ, Faber-Langendoen K, Jecker NS. Beyond futility to an ethic of care. *Am J Med.* 1994;96:110–114. [PubMed: 8109595]
-
161. Rocker G, Cook D, Sjøkvist P, et al. Clinician predictions of intensive care unit mortality. *Crit Care Med.* 2004;32:1149–1154. [PubMed: 15190965]
-
162. Curtis JR, Patrick DL, Shannon SE, et al. The family conference as a focus to improve communication about end-of-life care in the intensive care unit: opportunities for improvement. *Crit Care Med.* 2001;29:N26–N33.
-
163. Uhlmann RF, Pearlman RA. Perceived quality of life and preferences for life-sustaining treatment in older adults. *Arch Intern Med.* 1991;151:495–497. [PubMed: 2001131]
-
164. Kaplan SH, Greenfield S, Gandek B, et al. Characteristics of physicians with participatory decision-making styles. *Ann Intern Med.* 1996;124:497–504. [PubMed: 8602709]
-
165. Luce JM, Breeling JL. Critical care practices of chest physicians. *Chest.* 1988;93:163–165. [PubMed: 3335149]
-
166. Kelly WF, Eliasson AH, Stocker DJ, Hnatiuk OW. Do specialists differ on do-not-resuscitate decisions? *Chest.* 2002;121:957–963. [PubMed: 11888982]
-
167. Kollef MH. Private attending physician status and the withdrawal of life-sustaining interventions in a medical intensive care unit population. *Crit Care Med.* 1996;24:968–975. [PubMed: 8681600]
-
168. Azoulay E, Timsit JF, Sprung CL, et al. Prevalence and factors of intensive care unit conflicts: the conflicus study. *Am J Respir Crit Care Med.* 2009;180:853–860. [PubMed: 19644049]
-
169. Poncet MC, Toullic P, Papazian L, et al. Burnout syndrome in critical care nursing staff. *Am J Respir Crit Care Med.* 2007;175:698–704. [PubMed: 17110646]
-
170. Christakis NA, Asch DA. Biases in how physicians choose to withdraw life support. *Lancet.* 1993;342:642–646. [PubMed: 8103146]
-
171. Asch DA, Christakis NA. Why do physicians prefer to withdraw some forms of life support over others? Intrinsic attributes of life-sustaining treatments are associated with physicians' preferences. *Med Care.* 1996;34:103–111. [PubMed: 8632684]
-
172. Chelluri L, Im KA, Belle SH, et al. Long-term mortality and quality of life after prolonged mechanical ventilation. *Crit Care Med.* 2004;32:61–69. [PubMed: 14707560]
-
173. Herridge MS, Cheung AM, Tansey CM, et al. One-year outcomes in survivors of the acute respiratory distress syndrome. *N Engl J Med.* 2003;348:683–693. [PubMed: 12594312]
-
174. Im K, Belle SH, Schulz R, et al. Prevalence and outcomes of caregiving after prolonged mechanical ventilation in the ICU. *Chest.* 2004;125:597–606. [PubMed: 14769744]
-
175. Asch DA, Faber-Langendoen K, Shea JA, Christakis NA. The sequence of withdrawing life-sustaining treatment from patients. *Am J Med.* 1999;107:153–156. [PubMed: 10460047]

-
176. Cook DJ, Guyatt G, Rocker G, et al. Cardiopulmonary resuscitation directives on admission to intensive-care unit: an international observational study. *Lancet*. 2001;358:1941–1945. [[PubMed: 11747918](#)]
-
177. Walter SD, Cook DJ, Guyatt GH, et al. Confidence in life-support decisions in the intensive care unit: a survey of healthcare workers. Canadian Critical Care Trials Group. *Crit Care Med*. 1998;26:44–49. [[PubMed: 9428542](#)]
-
178. Azoulay E, Pochard F, Garrouste-Orgeas M, et al. Decisions to forgo life-sustaining therapy in ICU patients independently predict hospital death. *Intensive Care Med*. 2003;29:1895–1901. [[PubMed: 14530857](#)]
-
179. Shepardson LB, Youngner SJ, Speroff T, Rosenthal GE. Increased risk of death in patients with do-not-resuscitate orders. *Med Care*. 1999;37:727–737. [[PubMed: 10448716](#)]
-
180. Sulmasy DP. Do patients die because they have DNR orders, or do they have DNR orders because they are going to die? *Med Care*. 1999;37:719–721. [[PubMed: 10448714](#)]
-
181. White DB, Braddock CH 3rd, Bereksnyei S, Curtis JR. Toward shared decision making at the end of life in intensive care units: opportunities for improvement. *Arch Intern Med*. 2007;167:461–467. [[PubMed: 17353493](#)]
-
182. Luce JM. The art of negotiating. *Crit Care Med*. 2001;29:1078–1079. [[PubMed: 11383532](#)]
-
183. Rivera S, Kim D, Garone S, et al. Motivating factors in futile clinical interventions. *Chest*. 2001;119:1944–1947. [[PubMed: 11399728](#)]
-
184. Bowman KW. Communication, negotiation, and mediation: dealing with conflict in end-of-life decisions. *J Palliat Care*. 2000;16:S17–S23.
-
185. Prendergast TJ. Resolving conflicts surrounding end-of-life care. *New Horiz*. 1997;5:62–71. [[PubMed: 9017680](#)]
-
186. Miller DK, Coe RM, Hyers TM. Achieving consensus on withdrawing or withholding care for critically ill patients. *J Gen Intern Med*. 1992;7:475–480. [[PubMed: 1403201](#)]
-
187. Schneiderman LJ, Gilmer T, Teetzel HD, et al. Effect of ethics consultations on nonbeneficial life-sustaining treatments in the intensive care setting: a randomized controlled trial. *JAMA*. 2003;290:1166–1172. [[PubMed: 12952998](#)]
-
188. Westphal DM, McKee SA. End-of-life decision making in the intensive care unit: physician and nurse perspectives. *Am J Med Qual*. 2009; 24:222–228. [[PubMed: 19372540](#)]
-
189. Sprung CL, Carmel S, Sjobqvist P, et al. Attitudes of European physicians, nurses, patients, and families regarding end-of-life decisions: the ETHICATT study. *Intensive Care Med*. 2007;33:104–110. [[PubMed: 17066284](#)]
-
190. White DB, Curtis JR, Lo B, Luce JM. Decisions to limit life-sustaining treatment for critically ill patients who lack both decision-making capacity and surrogate decision-makers. *Crit Care Med*. 2006;34:2053–2059. [[PubMed: 16763515](#)]
-
191. Mularski RA, Bascom P, Osborne ML. Educational agendas for interdisciplinary end-of-life curricula. *Crit Care Med*. 2001;29:N16–N23.
-

192. Clarke EB, Curtis JR, Luce JM, et al. Quality indicators for end-of-life care in the intensive care unit. *Crit Care Med*. 2003;31:2255–2262. [[PubMed: 14501954](#)]
-
193. Johnson N, Cook D, Giacomini M, Willms D. Towards a “good” death: end-of-life narratives constructed in an intensive care unit. *Cult Med Psychiatry*. 2000;24:275–295. [[PubMed: 11012101](#)]
-
194. Fins JJ, Solomon MZ. Communication in intensive care settings: the challenge of futility disputes. *Crit Care Med*. 2001;29:N10–N15.
-
195. Lilly CM, Sonna LA, Haley KJ, Massaro AF. Intensive communication: four-year follow-up from a clinical practice study. *Crit Care Med*. 2003;31:S394–S399.
-
196. Seymour JE. Negotiating natural death in intensive care. *Soc Sci Med*. 2000;51:1241–1252. [[PubMed: 11037214](#)]
-
197. Cook DJ, Giacomini M, Johnson N, Willms D. Life support in the intensive care unit: a qualitative investigation of technological purposes. Canadian Critical Care Trials Group. *CMAJ*. 1999;161:1109–1113. [[PubMed: 10569086](#)]
-
198. Curtis JR, Patrick DL, Engelberg RA, et al. A measure of the quality of dying and death. Initial validation using after-death interviews with family members. *J Pain Symptom Manage*. 2002;24:17–31. [[PubMed: 12183092](#)]
-
199. Fried TR, Bradley EH, Towle VR, Allore H. Understanding the treatment preferences of seriously ill patients. *N Engl J Med*. 2002;346:1061–1066. [[PubMed: 11932474](#)]
-
200. Curtis JR, Engelberg RA, Wenrich MD, et al. Missed opportunities during family conferences about end-of-life care in the intensive care unit. *Am J Respir Crit Care Med*. 2005;171:844–849. [[PubMed: 15640361](#)]
-
201. Danis M, Federman D, Fins JJ, et al. Incorporating palliative care into critical care education: principles, challenges, and opportunities. *Crit Care Med*. 1999;27:2005–2013. [[PubMed: 10507632](#)]
-
202. DeVita MA, Arnold RM, Barnard D. Teaching palliative care to critical care medicine trainees. *Crit Care Med*. 2003;31:1257–1262. [[PubMed: 12682501](#)]
-
203. Nyman DJ, Sprung CL. End-of-life decision making in the intensive care unit. *Intensive Care Med*. 2000;26:1414–1420. [[PubMed: 11126250](#)]
-
204. Nelson JE, Danis M. End-of-life care in the intensive care unit: where are we now? *Crit Care Med*. 2001;29:N2–N9.
-
205. Curtis JR, Engelberg RA, Wenrich MD, et al. Studying communication about end-of-life care during the ICU family conference: development of a framework. *J Crit Care*. 2002;17:147–160. [[PubMed: 12297990](#)]
-
206. Gries CJ, Curtis JR, Wall RJ, Engelberg RA. Family member satisfaction with end-of-life decision making in the ICU. *Chest*. 2008;133:704–712. [[PubMed: 18198256](#)]
-
207. Selph RB, Shiang J, Engelberg R, et al. Empathy and life support decisions in intensive care units. *J Gen Intern Med*. 2008;23:1311–1317. [[PubMed: 18574641](#)]
-

208. Keenan SP, Mawdsley C, Plotkin D, et al. Withdrawal of life support: how the family feels, and why. *J Palliat Care*. 2000;16:S40–S44.
-
209. Steinhauser KE, Christakis NA, Clipp EC, et al. Factors considered important at the end of life by patients, family, physicians, and other care providers. *JAMA*. 2000;284:2476–2482. [[PubMed: 11074777](#)]
-
210. Robinson SM, Mackenzie-Ross S, Campbell Hewson GL, et al. Psychological effect of witnessed resuscitation on bereaved relatives. *Lancet*. 1998;352:614–617. [[PubMed: 9746023](#)]
-
211. McDonagh JR, Elliott TB, Engelberg RA, et al. Family satisfaction with family conferences about end-of-life care in the intensive care unit: increased proportion of family speech is associated with increased satisfaction. *Crit Care Med*. 2004;32:1484–1488. [[PubMed: 15241092](#)]
-
212. Fridh I, Forsberg A, Bergbom I. Family presence and environmental factors at the time of a patient's death in an ICU. *Acta Anaesthesiol Scand*. 2007;51:395–401. [[PubMed: 17378776](#)]
-
213. Lo B, Ruston D, Kates LW, et al. Discussing religious and spiritual issues at the end of life: a practical guide for physicians. *JAMA*. 2002;287:749–754. [[PubMed: 11851542](#)]
-
214. Wall RJ, Engelberg RA, Gries CJ, et al. Spiritual care of families in the intensive care unit. *Crit Care Med*. 2007;35:1084–1090. [[PubMed: 17334245](#)]
-
215. Lautrette A, Darmon M, Megarbane B, et al. A communication strategy and brochure for relatives of patients dying in the ICU. *N Engl J Med*. 2007;356:469–478. [[PubMed: 17267907](#)]
-
216. Asch DA. The role of critical care nurses in euthanasia and assisted suicide. *N Engl J Med*. 1996;334:1374–1379. [[PubMed: 8614424](#)]
-
217. Ferrand E, Lemaire F, Regnier B, et al. Discrepancies between perceptions by physicians and nursing staff of intensive care unit end-of-life decisions. *Am J Respir Crit Care Med*. 2003;167:1310–1315. [[PubMed: 12738597](#)]
-
218. Puntillo KA, Benner P, Drought T, et al. End-of-life issues in intensive care units: a national random survey of nurses' knowledge and beliefs. *Am J Crit Care*. 2001;10:216–229. [[PubMed: 11432210](#)]
-
219. Kirchhoff KT, Beckstrand RL. Critical care nurses' perceptions of obstacles and helpful behaviors in providing end-of-life care to dying patients. *Am J Crit Care*. 2000;9:96–105. [[PubMed: 10705423](#)]
-
220. Stevens L, Cook D, Guyatt G, et al. Education, ethics, and end-of-life decisions in the intensive care unit. *Crit Care Med*. 2002;30:290–296. [[PubMed: 11889295](#)]
-
221. Moreau D, Goldgran-Toledano D, Alberti C, et al. Junior versus senior physicians for informing families of intensive care unit patients. *Am J Respir Crit Care Med*. 2003;4:4.
-
222. Machare Delgado E, Callahan A, Paganelli G, et al. Multidisciplinary family meetings in the ICU facilitate end-of-life decision making. *Am J Hosp Palliat Care*. 2009;26:295–302.
-

223. Bradley CT, Webb TP, Schmitz CC, et al. Structured teaching versus experiential learning of palliative care for surgical residents. *Am J Surg*. 2010;200:542–547. [PubMed: 20538256]
-
224. Cheung W, Aggarwal G, Fugaccia E, et al. Palliative care teams in the intensive care unit: a randomised, controlled, feasibility study. *Crit Care Resusc*. 2010;12:28–35. [PubMed: 20196711]
-
225. Kaufer M, Murphy P, Barker K, Mosenthal A. Family satisfaction following the death of a loved one in an inner city MICU. *Am J Hosp Palliat Care*. 2008;25:318–325. [PubMed: 18539762]
-
226. Mosenthal AC, Murphy PA, Barker LK, et al. Changing the culture around end-of-life care in the trauma intensive care unit. *J Trauma*. 2008;64:1587–1593. [PubMed: 18545128]
-
227. Kuschner WG, Gruenewald DA, Clum N, et al. Implementation of ICU palliative care guidelines and procedures: a quality improvement initiative following an investigation of alleged euthanasia. *Chest*. 2009;135:26–32. [PubMed: 19136403]
-
228. Gebara J, Tashjian H. End-of-life practices at a Lebanese hospital: courage or knowledge? *J Transcult Nurs*. 2006;17:381–388. [PubMed: 16946121]
-
229. Curtis JR, Patrick DL, Caldwell ES, Collier AC. Why don't patients and physicians talk about end-of-life care? Barriers to communication for patients with acquired immunodeficiency syndrome and their primary care clinicians. *Arch Intern Med*. 2000;160:1690–1696. [PubMed: 10847263]
-
230. Hanson LC, Tulsky JA, Danis M. Can clinical interventions change care at the end of life? *Ann Intern Med*. 1997;126:381–388. [PubMed: 9054283]
-
231. Treece PD, Engelberg RA, Crowley L, et al. Evaluation of a standardized order form for the withdrawal of life support in the intensive care unit. *Crit Care Med*. 2004;32:1141–1148. [PubMed: 15190964]
-
232. Arnold RM, Kellum J. Moral justifications for surrogate decision making in the intensive care unit: implications and limitations. *Crit Care Med*. 2003;31:S347–S353.
-
233. Goold SD, Williams B, Arnold RM. Conflicts regarding decisions to limit treatment: a differential diagnosis. *JAMA*. 2000;283:909–914. [PubMed: 10685716]
-
234. Hammes BJ, Rooney BL. Death and end-of-life planning in one midwestern community. *Arch Intern Med*. 1998;158:383–390. [PubMed: 9487236]
-
235. Nelson JE, Angus DC, Weissfeld LA, et al. End-of-life care for the critically ill: a national intensive care unit survey. *Crit Care Med*. 2006;34:2547–2553. [PubMed: 16932230]
-
236. Shanawani H, Wenrich MD, Tonelli MR, Curtis JR. Meeting physicians' responsibilities in providing end-of-life care. *Chest*. 2008;133:775–786. [PubMed: 18321905]
-
237. Walling AM, Asch SM, Lorenz KA, et al. The quality of care provided to hospitalized patients at the end of life. *Arch Intern Med*. 2010;170:1057–1063. [PubMed: 20585072]
-

238. White DB, Engelberg RA, Wenrich MD, et al. Prognostication during physician-family discussions about limiting life support in intensive care units. *Crit Care Med*. 2007;35:442–448. [[PubMed: 17205000](#)]

239. Mohan D, Alexander SC, Garrigues SK, et al. Communication practices in physician decision-making for an unstable critically ill patient with end-stage cancer. *J Palliat Med*. 2010;13:949–956. [[PubMed: 20642362](#)]

240. White DB, Malvar G, Karr J, et al. Expanding the paradigm of the physician's role in surrogate decision-making: an empirically derived framework. *Crit Care Med*. 2010;38:743–750. [[PubMed: 20029347](#)]

241. Thornton JD, Pham K, Engelberg RA, et al. Families with limited English proficiency receive less information and support in interpreted intensive care unit family conferences. *Crit Care Med*. 2009;37:89–95. [[PubMed: 19050633](#)]

242. Campbell ML, Guzman JA. A proactive approach to improve end-of-life care in a medical intensive care unit for patients with terminal dementia. *Crit Care Med*. 2004;32:1839–1843. [[PubMed: 15343010](#)]

243. Campbell ML, Guzman JA. Impact of a proactive approach to improve end-of-life care in a medical ICU. *Chest*. 2003;123:266–271. [[PubMed: 12527629](#)]

[McGraw Hill](#)

Copyright © McGraw-Hill Global Education Holdings, LLC.

All rights reserved.

Your IP address is **132.174.254.28**

Access Provided by: Walter Reed National Military Medical Center

[Silverchair](#)